Lecture 10 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a program that asks the user to enter there age, and then tells them if they are old enough to vote in the BC provincial election. You must be 18 or older to vote in BC.

Sample run 1

```
Enter your age: 18
You are old enough to vote in the BC provincial election.
```

Sample run 2

```
Enter your age: 15
You are not old enough to vote in the BC provincial election.
```

Sample Solution

```
age = int(input("Enter your age: "))
if age >= 18:
    print("You are old enough to vote in the BC provincial election.")
else:
    print("You are not old enough to vote in the BC provincial election.")
```

Question 2 (q2_sol.py)

Write a program that asks the user to enter the temperature and then tells them what to wear according to these rules:

- If the temperature is 10 degrees or less, they should wear a warm coat.
- If the temperature is greater than 10 degrees and less than 25 degrees, they should wear a light jacket.
- If the temperature is greater than 25 degrees, they should wear a t-shirt.

Sample run 1

```
Enter the temperature: 10
Wear a warm coat.
```

Sample run 2

```
Enter the temperature: 20
Wear a light jacket.
```

Sample run 3

```
Enter the temperature: 31.5
Wear a t-shirt.
```

Sample Solution

```
temp = float(input("Enter the temperature: "))
if temp <= 10:
    print("Wear a warm coat.")
elif 10 < temp < 25:
    print("Wear a light jacket.")
else:
    print("Wear a t-shirt.")
```

Question 3 (q3_sol.py)

The community pool charges $8 for admission, unless you fall into one of these categories:

- Seniors, aged 65 or older, pay $5
- Children, aged 6 through 12, pay $4
- Kids under 6 get in free

Write a program that asks the user their age and then prints their price of admission.

Sample run 1

```
Enter your age: 65
Admission: $5
```

Sample run 2

```
Enter your age: 22
Admission: $8
```

Sample run 3

```
Enter your age: 5
Admission: free!
```

Sample Solution

```
age = int(input("Enter your age: "))
if age >= 65:
    print("Admission: $5")
elif 13 <= age <= 64:
    print("Admission: $8")
elif 6 <= age <= 12:
    print("Admission: $4")
else:
    print("Admission: free!")
```

Question 4 (q4_sol.py)

During the summer, a neighborhood has these rules for lawn watering:

- houses with even street numbers may water their lawn only on even-numbered days
- houses with odd street numbers may water their lawn only on odd-numbered days

Write a program that asks the user to enter a street number (like 503 or 1944) and a day number (Monday is 1, Tuesday is 2, ..., Sunday is 7), and then prints whether or not that house may water their lawn that day.

As an extra challenge, write and use a function called `can_water(street_number, day)` that returns `True` if the house may water their lawn that day, and `False` otherwise.

Sample run 1

```
Enter a street number: 503
Enter a day number: 1
You can water your lawn today.
```

Sample run 2

```
Enter a street number: 1944
Enter a day number: 2
You can water your lawn today.
```

Sample run 3

```
Enter a street number: 1944
Enter a day number: 3
You cannot water your lawn today.
```

Sample Solution

```
def can_water(street_number, day):
    if street_number % 2 == 0:
        return day % 2 == 0
    else:
        return day % 2 != 0

street_number = int(input("Enter a street number: "))
day = int(input("Enter a day number: "))
if can_water(street_number, day):
    print("You can water your lawn today.")
else:
    print("You cannot water your lawn today.")
```